Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER B.TECH. (ELECTRICAL &ELECTRONICS)

END SEMESTER EXAMINATIONS, MAY 2024

DATA ANALYTICS [ELE 4077] PROGRAM ELECTIVE VI

REVISED CREDIT SYSTEM

Time: 3 Hours Date: MAY 2024 Max. Marks: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. List the different methods of data analysis? ***(02)***

1B. Develop a Python program to identify and display all prime numbers within a specified range? ***(03)***

1C. Test scores for a college statistics class held during the day are:

99; 56; 78; 55.5; 32; 90; 80; 81; 56; 59; 45; 77; 84.5; 84; 70; 72; 68; 32; 79; 90

Test scores for a college statistics class held during the evening class are:

98; 78; 68; 83; 81; 89; 88; 76; 65; 45; 98; 90; 80; 84.5; 85; 79; 78; 98; 90; 79; 81; 25.5

(i) Find the smallest and largest values, the median, and the first and third quartile for the day class and evening class.

(ii) Create a box plot for each set of data. Use one number line for both box plots.

(iii) Which box plot has the widest spread for the middle 50% of the data (the data between the first and third quartiles)? What does this mean for that set of data in comparison to the other set of data? ***(05)***

2A. Solve for the Coefficient of Skewness and Coefficient of Kurtosis for the given data set and interpret the results obtained?

X={3,6,7,7,8,10,12,14,18,20} ***(03)***

2B. Below are the details regarding the class of defective bulbs and the number of boxes. Evaluate the mean, median, and mode for the given grouped data set.

Class No. of defective bulbs	No. of boxes (f)
0-2	3
2-4	4
4-6	5
6-8	3
8-10	1
Total	16

(04)

2C. Describe the role of the Autocorrelation function (ACF) in time series analysis. From Fig. (a) and Fig. (b), interpret the time series data trend, autocorrelation pattern and the order of the autoregression moving average model (ARMA).

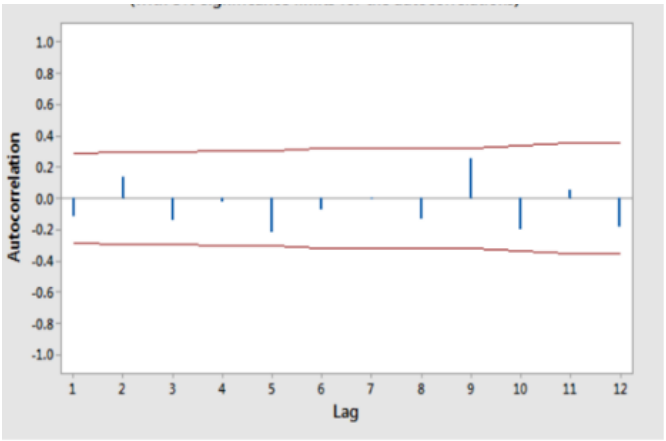


Fig.(a)

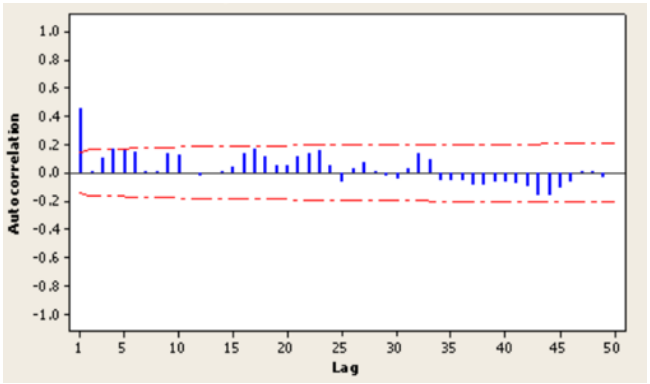


Fig. (b)

(03)

3A. Define stationarity in time series of data and explain Augmented Dickey-Fuller (ADF) Test. ***(02)***

3B. The following data indicates the annual sales (in millions) for a company over 10 years. Evaluate the 5-year moving average for this sales data and comment on the trend in sales over the years.

Year	**Sales (in millions)**
2013	4
2014	6
2015	5
2016	8
2017	9
2018	5
2019	4
2020	3
2021	7
2022	8

(03)

3C. A botanist wants to explore how sunlight exposure and watering frequency affect plant growth. She plants 40 seeds and lets them grow for two months under different conditions for sunlight exposure and watering frequency. After two months, she records the height of each plant. Using the data given, apply Two Way ANOVA test to determine at 5% significance level, does sunlight exposure and watering frequency affect plant growth? Also determine if there is an interaction effect between sunlight exposure and watering frequency. Note: Fcritical value at α=0.05 significance level from the F distribution table is for sunlight and watering are 2.90112 and 4.149097 respectively. Similarly, Fcritical value at α=0.05 significance level from the F distribution table is for interaction effect is 2.90112.

	Sunlight Exposure			
Watering Frequency	None	Low	Medium	High
Daily	4.8	5	6.4	6.3
	4.4	5.2	6.2	6.4
	3.2	5.6	4.7	5.6
	3.9	4.3	5.5	4.8
	4.4	4.8	5.8	5.8
Weekly	4.4	4.9	5.8	6
	4.2	5.3	6.2	4.9
	3.8	5.7	6.3	4.6
	3.7	5.4	6.5	5.6
	3.9	4.8	5.5	5.5

(05)

4A. Construct the regression equation of Y on X (where Y is the selling price and X is the square footage) and hence use the regression model to estimate the price of a house with 2,200 square feet. Also Calculate the following performance metrics: Mean Absolute Error (MAE), R-squared (R^2) to evaluate the model's accuracy.

Square footage (X)	**Selling Price (Y in $ 1000)**
1000	150
1500	200
2000	250
2500	300
3000	350

(04)

4B. Two teaching methods are tested to see if they result in different average scores. Method 1 is tested on 30 students with an average score of 82 and a standard deviation of 10, while Method 2 is tested on 35 students with an average score of 85 and a standard deviation of 12. Is there a significant difference between the two methods at the 0.05 significance level? (Apply Z test to evaluate difference in teaching method).

Note: For a two-tailed test at the 0.05 significance level, the critical Z-values are approximately ±1.96. ***(03)***

4C. Two brands of tires are tested for durability. At 5% significance level, can we conclude that two brands have different average life spans. [Note: Perform T test to find the significant difference between the mean scores].

Brand A has the following life spans (in miles)
40,000; 42,000; 41,500; 43,000; 39,500. ***(03)***

Brand B has the following life spans (in miles)
38,000; 37,500; 39,000; 38,500; 37,000.

Note: For a two-tailed test at the 5% significance level α=0.05 with df=8, the critical t-value is approximately 2.306.

5A. Perform Chi-Square Test at 5% significance level for Independence on the following data to determine if an association (relationship) exists between marital status and educational qualification. From X^2 distribution table, for df $=$ 12 at 0.05 significance level is $= 21.03$

Qualification/Marital Status	Middle School	High School	Bachelor's	Master's	Ph.D
Never married	18	36	21	9	6
Married	12	36	45	36	21
Divorced	6	9	9	3	3
Windowed	3	9	9	6	3

(04)

5B. List any six differences between the two main areas of statistics, descriptive and inferential statistics. ***(03)***

5C. The following table gives the score of 5 students in Mathematics and Computer Science in the first internal assessment test. Evaluate the Spearman's Rank Correlation for the given data and hence comment on the correlation between two ranks.

Students	Mathematics	Computer Science
A	35	24
B	20	35
C	49	39
D	44	48
E	30	45

(03)